ELK406E
Distribution of Electrical Energy

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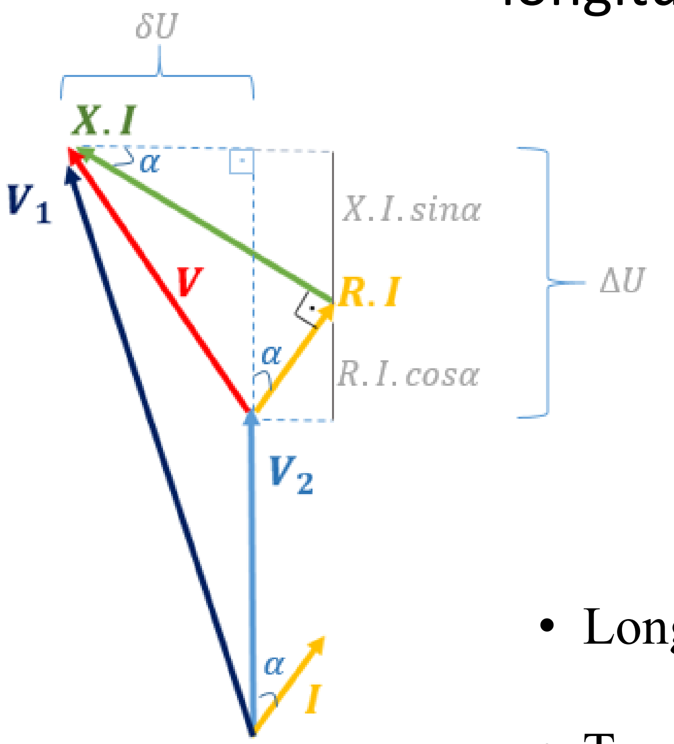
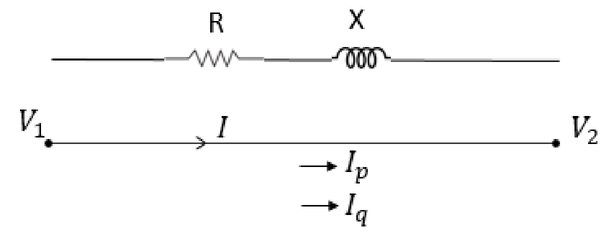
Voltage Drop

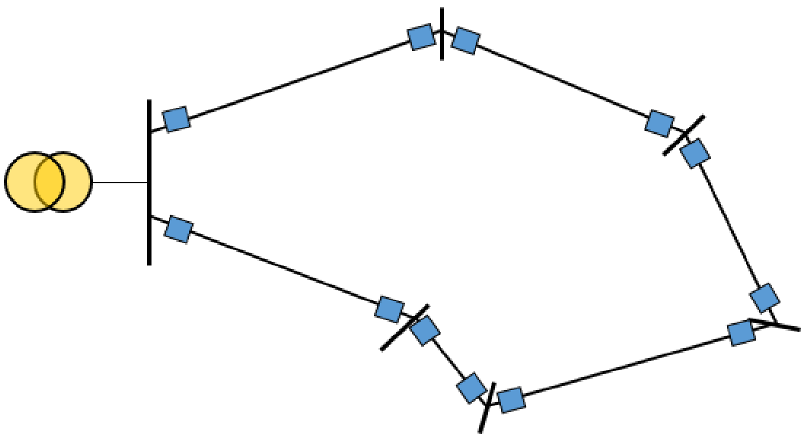


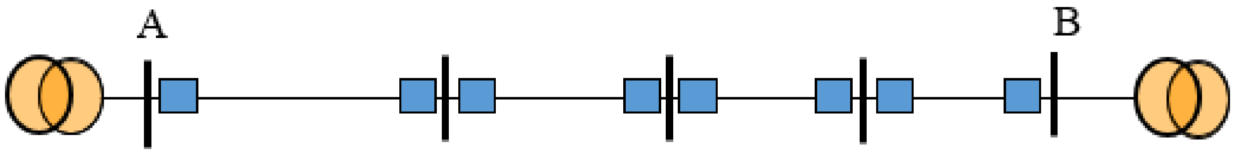
- Voltage drop has two components, a longitudinal and a transversal component



- Longitudinal $Voltage\ Drop,\ \Delta U = R\ I_p + X\ I_q$
- Transversal $Voltage\ Drop,\ \delta U = X\ I_p - R\ I_q$

Closed Loop VD

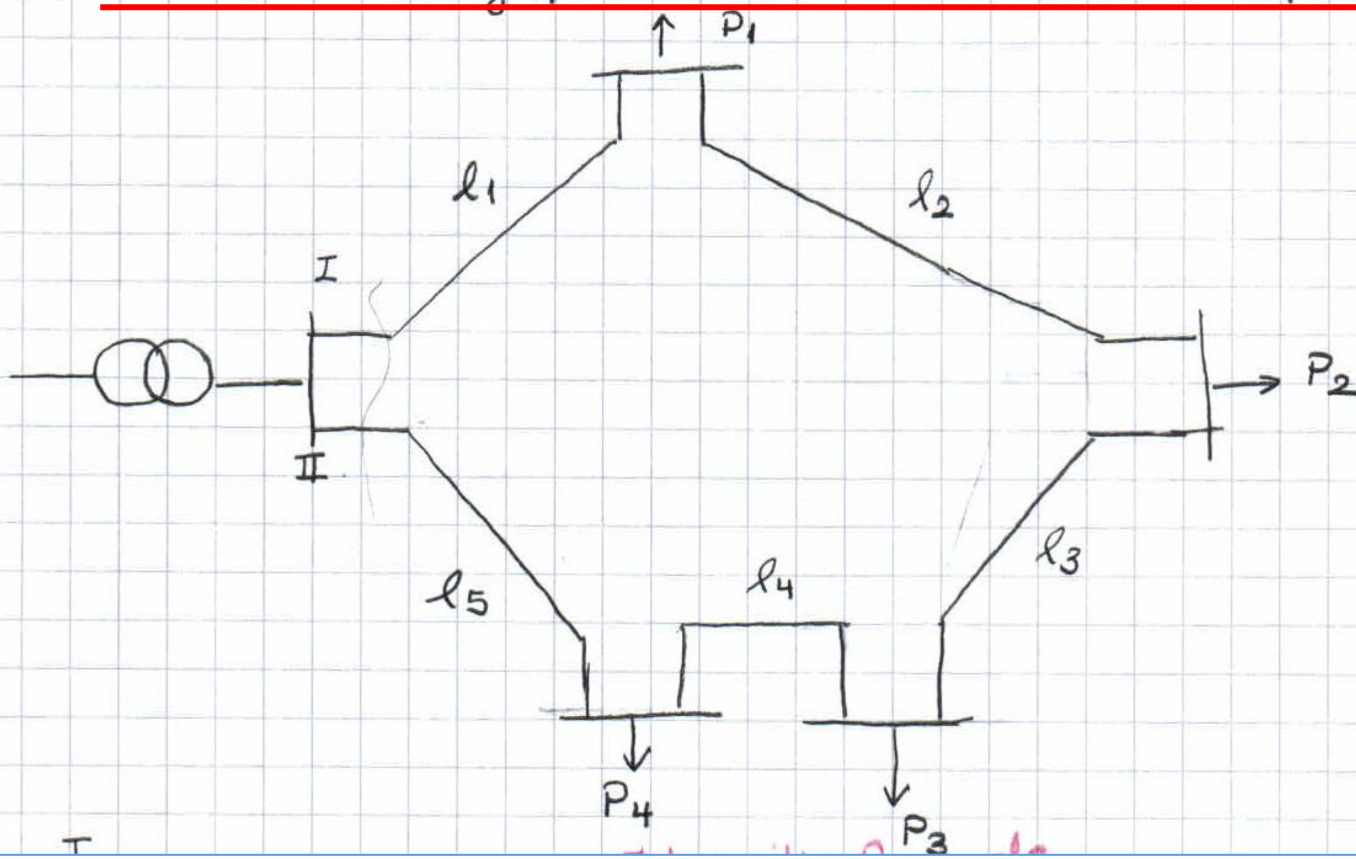


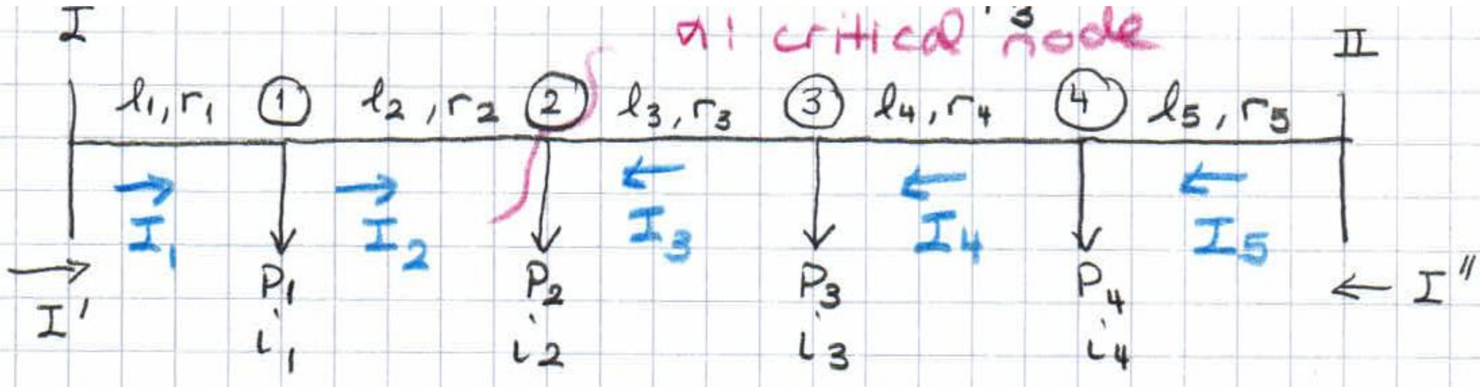


Closed loop systems

In closed loop systems, the feeders are fed at both ends.

- Feeders being fed at both ends with equal voltage.

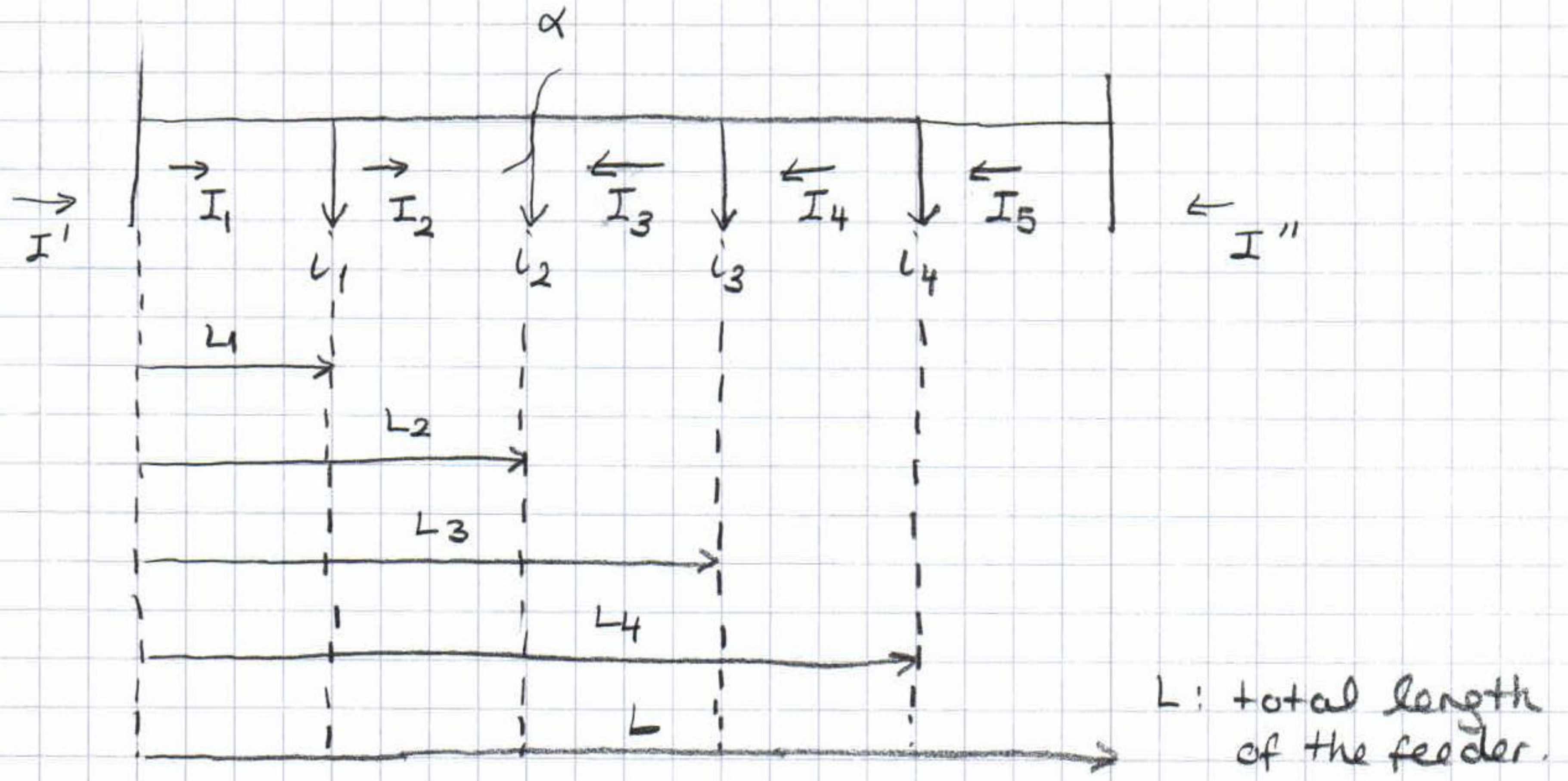




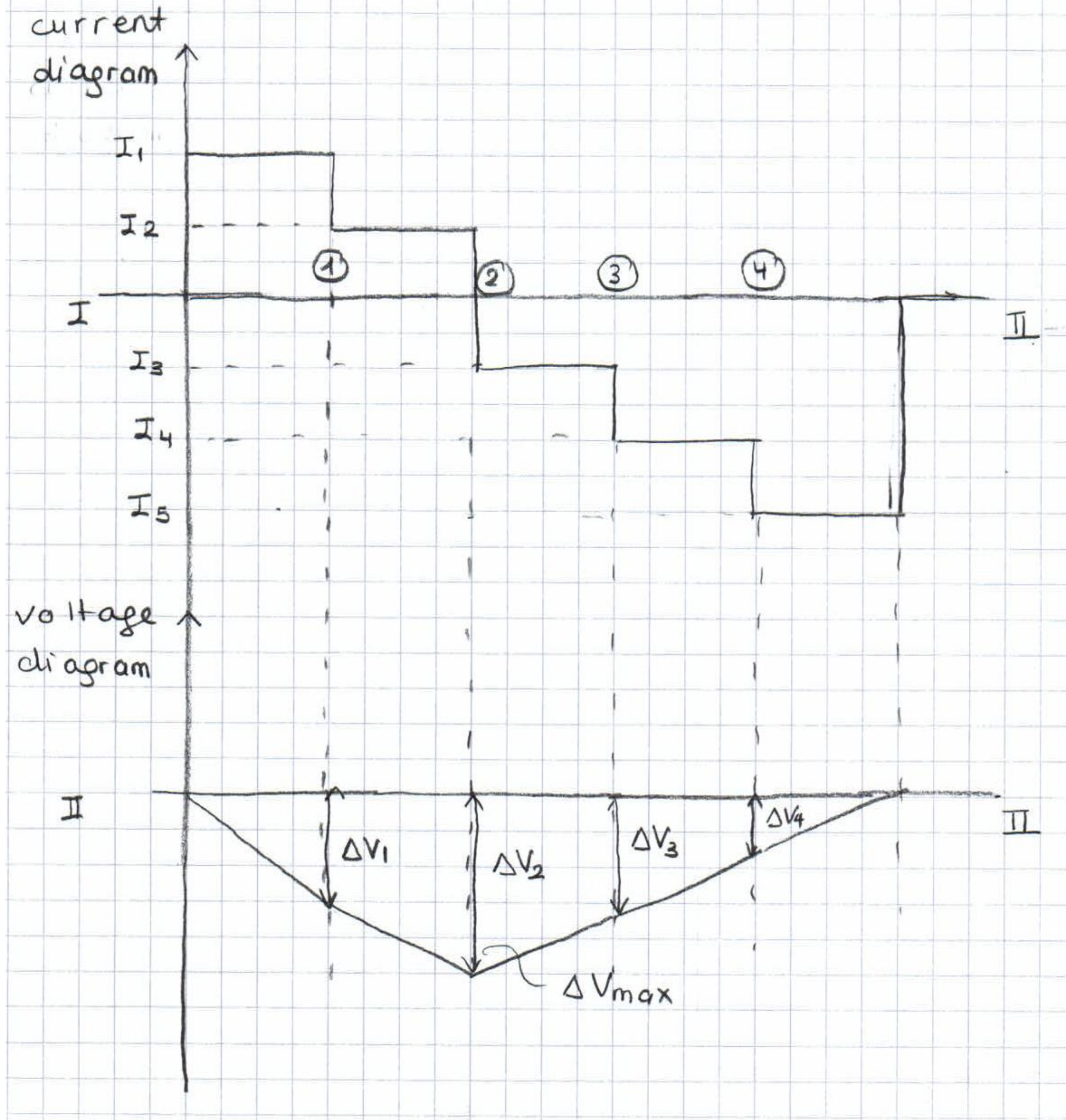
I': current supplied from I.

I'': current supplied from II

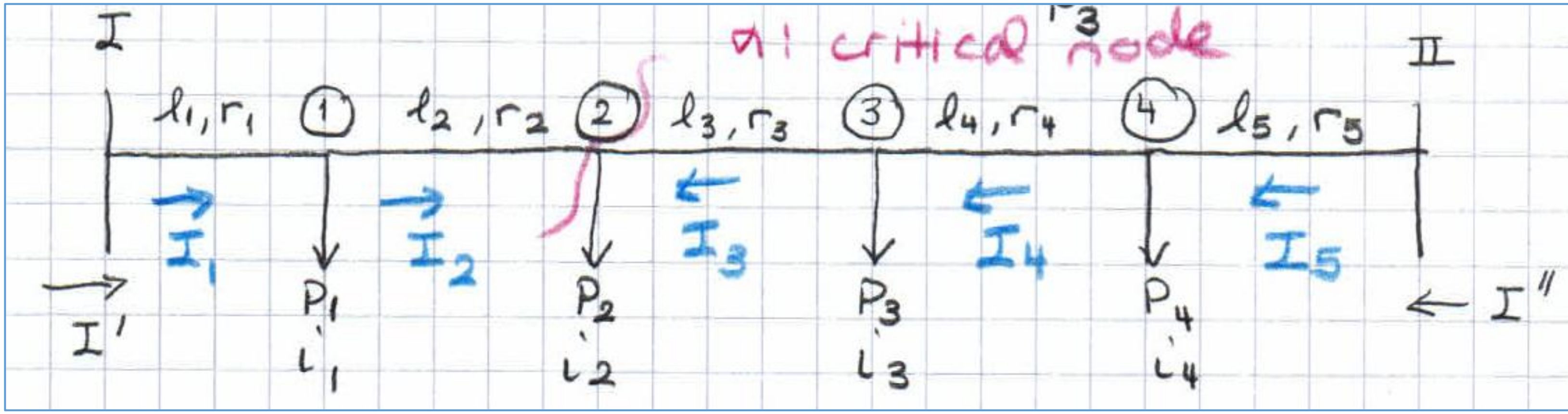
Since $V_I = V_{II}$, total voltage drop along the feeder is zero. ($\Delta V = 0$). That means, there's a critical node α where the potential is minimum or in other words, the voltage drop is maximum. At the critical node α, the line current changes direction.



α
I'
I₁
I₂
I₃
I₄
I₅
I''
i₁
i₂
i₃
i₄
L₁
L₂
L₃
L₄
L
L: total length of the feeder.



current diagram
I1
I2
1
2
3
4
I
II
I3
I4
I5
voltage diagram
II
ΔV1
ΔV2
ΔV3
ΔV4
II
ΔVmax



The aim is to find the current disribution (line currents) along the feeder and the critical node

Total voltage drop along the feeder;

$$r_1 I_1 + r_2 I_2 + r_3 I_3 + r_4 I_4 + r_5 I_5 = 0$$

Let's express each line current in terms of I_5.

$$r_1(I_5 - i_4 - i_3 - i_2 - i_1) + r_2(I_5 - i_4 - i_3 - i_2) + r_3(I_5 - i_4 - i_3)$$

$$+ r_4(I_5 - i_4) + r_5 I_5 = 0$$

$$I_5 \underbrace{(r_1 + r_2 + r_3 + r_4 + r_5)}_{L \cdot r} = i_4 \underbrace{(r_1 + r_2 + r_3 + r_4)}_{L_4 \cdot r} + i_3 \underbrace{(r_1 + r_2 + r_3)}_{L_3 \cdot r}$$

$$+ i_2 \underbrace{(r_1 + r_2)}_{L_2 \cdot r} + \underbrace{r_1}_{L_1 \cdot r} i_1$$

r : resistance per unit length

$$I_5 \cdot L \cdot \cancel{r} = i_4 \cdot L_4 \cdot r + i_3 L_3 \cdot \cancel{r} + i_2 L_2 \cdot \cancel{r} + i_1 L_1 \cdot \cancel{r}$$

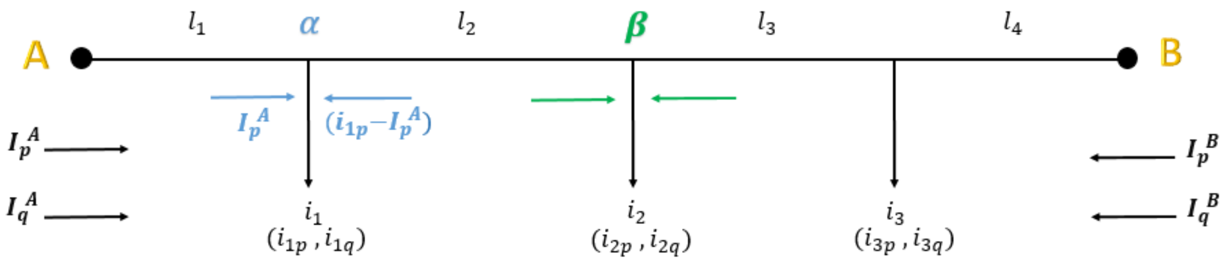
$$I_5 \cdot L = i_4 \cdot L_4 + i_3 L_3 + i_2 L_2 + i_1 L_1$$

$$= \sum_{n=1}^{4} i_n L_n$$: sum of the moments of load currents about feeding pt. I.

$$I'' = I_5 = \frac{\sum_{n=1}^{4} i_n L_n}{L}$$

$$\Delta V_{I-\alpha} = \Delta V_{II-\alpha}$$

Closed Loop VD

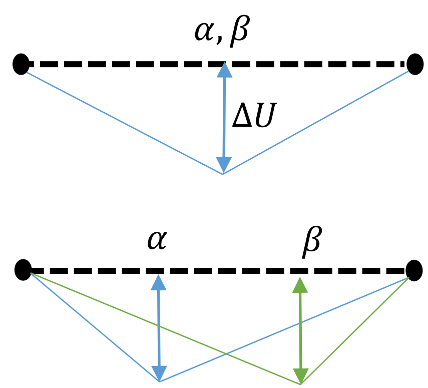


$$I_p{}^B = \frac{(i_{1p}\times l_1) + i_{2p}\times(l_1+l_2) + i_{3p}\times(l_1+l_2+l_3)}{l_1+l_2+l_3+l_4}$$

$$I_q{}^B = \frac{(i_{1q}\times l_1) + i_{2q}\times(l_1+l_2) + i_{3q}\times(l_1+l_2+l_3)}{l_1+l_2+l_3+l_4}$$

$$I_p{}^A = (i_{1p} + i_{2p} + i_{3p}) - I_p{}^B$$

$$I_q{}^A = (i_{1q} + i_{2q} + i_{3q}) - I_q{}^B$$

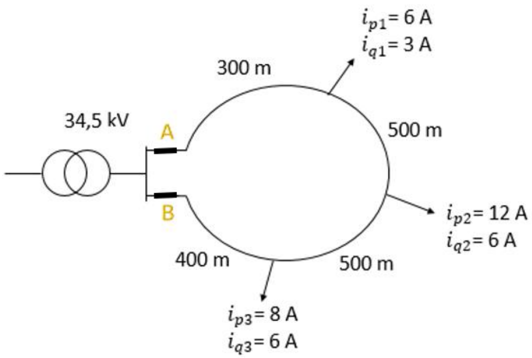


$\Delta U_{\ max} \rightarrow \alpha, \beta$

$\Delta U_{\ max} \rightarrow \alpha$ veya β

α: The node where the direction of active current flow changes
β: The node where the direction of reactive current flow changes

Örnek



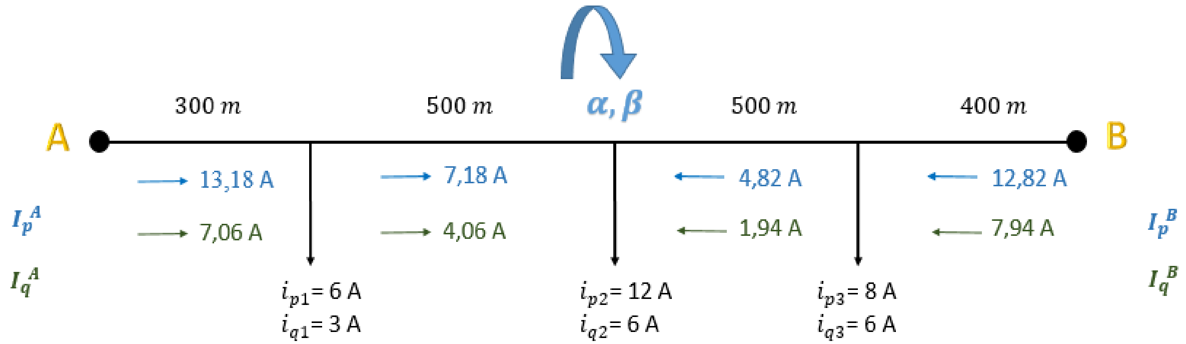
a) Calculate maximum VD.

b) Calculate total losses.

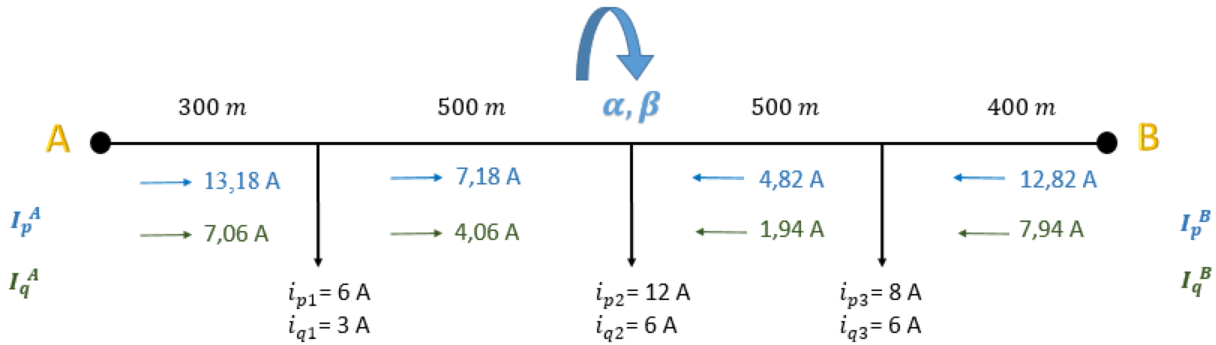
$Ҡ = 44{,}4\ m/\Omega mm^2$

$q = 35\ mm^2$

$L = 0{,}51\ mH/km$



- $I_p{}^B = \frac{\sum_{i=1}^{L} L_i \times i_{pi}}{L} = \frac{(300\times6)+(800\times12)+(1300\times8)}{1700} = 12{,}82\ A$
- $I_p{}^A$= (6+12+8)-12,82 = 13,18 A
- $I_q{}^B = \frac{\sum_{i=1}^{L} L_i \times i_{qi}}{L} = \frac{(300\times3)+(800\times6)+(1300\times6)}{1700}$ = 7,94 A
- $I_q{}^A$ = (3+6+6)-7,94= 7,06 A



$$\boldsymbol{X' = \omega} \times \boldsymbol{L} = 2 \times \pi \times f \times L = 2 \times \pi \times 50 \times \frac{0{,}51}{1000} = 0{,}16\,\frac{\Omega}{km}$$

$$\mathbf{X} = \boldsymbol{X'} \times \boldsymbol{l}$$

$$\Delta \boldsymbol{U} = \boldsymbol{R}\,\boldsymbol{I_p} + \boldsymbol{X}\,\boldsymbol{I_q} = \frac{1}{44{,}4 \times 35}(300 \times 13{,}18 + 500 \times 7{,}18) + (0{,}16 \times 0{,}3 \times 7{,}06) + (0{,}16 \times 0{,}5 \times 4{,}06) = 5{,}52\,V$$

$$\boldsymbol{\delta U} = \boldsymbol{X}\,\boldsymbol{I_p} - \boldsymbol{R}\,\boldsymbol{I_q} = (0{,}16 \times 0{,}3 \times 13{,}18) + (0{,}16 \times 0{,}5 \times 7{,}18) - \frac{1}{Ꝁ.q}(300 \times 7{,}06 + 500 \times 4{,}06) = -1{,}46\,V$$

$$\sqrt{\Delta U^2 + \delta U^2} = 5{,}7\,V\ (VDmax)$$

$$\boldsymbol{P_k} = \mathbf{3}\,\boldsymbol{R}\,\boldsymbol{I^2} = 3\,\frac{l}{Ꝁ.\,q}\left(i_p^2 + i_q^2\right) = 397\,W \qquad (total\ losses)$$

REFERENCES

- T. Gönen “Electric Power Distriburion System Engineering”
- Venkata, Lecture Notes
- Electric Power Distribution Handbook